

FaceRankNet: An Interpretable Part-Based Graph Attention Network for Geometry-Driven Facial Beauty Prediction

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1. Motivation & Problem Statement

Traditional CNN-based FBP models demonstrate high predictive power but lack an essential characteristic of being a “black box.” They act as a black-box predictor producing an estimate of general beauty regardless of what exact facial features contribute to it. The post-hoc analysis techniques (saliency maps), although useful for understanding how the model works, require training beforehand and can be influenced by demographic factors, such as different skin tones and illumination. FaceRankNet tackles this problem by constructing a geometry-based graph attention network.

2. Method: FaceRankNet Architecture

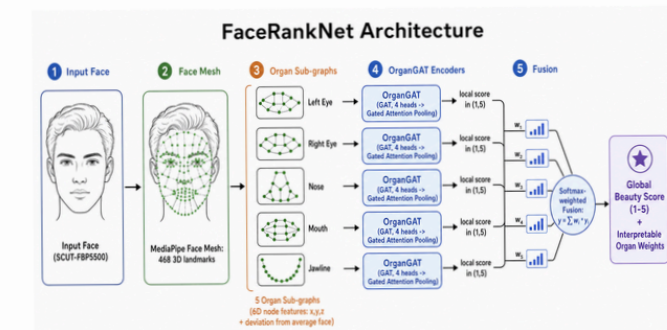


Figure 1: The FaceRankNet pipeline from facial mesh extraction to interpretable global score fusion.

- Geometry Only Inputs:** This model detects 468 three-dimensional points using the MediaPipe Face Mesh technique, with normalization based on the inter-ocular distance. RGB point cloud information is ignored to explicitly reduce biases due to demographics in the input layer.
- Node Feature Information:** Each node has six features represented as $(x, y, z, \Delta x, \Delta y, \Delta z)$, wherein Δ is the geometric distance from the Universal Average Face.
- Organ Sub-Graphs:** The face is decomposed into five organs through subgraphs that represent the left eye, right eye, nose, mouth, and jawline.
- OrganGAT Encoder:** Every organ subgraph passes through a unique Graph Attention Network (GAT) consisting of four attention heads and gated attention pooling that provides the local score of each organ, $\hat{y}_{organ_i} \in [1, 5]$
- Model Interpretation Through Fusion:** Using a trainable mechanism known as Softmax, the model fuses the local scores of each organ to make the final prediction, thus providing the relative importance of each organ to overall beauty:

$$\hat{y}_{global} = \sum_{i=1}^5 softmax(w_i) \hat{y}_{organ_i} \in [1, 5]$$

3. Weak Supervision: Beauty-Axis Projection

In order to deal with the total absence of any attractiveness annotations at the local part level, pseudo-labels for each organ are generated using projections onto an aesthetics direction learned specifically for each ethnicity (Hypothesis H1). Instead of considering symmetry distance to the centroid, a directional vector is introduced, which runs from the mean of the group μ_e to an exemplar in the top-30% most attractive individuals μ_e , where $a_e = \pi_e - \mu_e$. The resulting projections are mapped quantiles, producing a robust pseudo-score range $[1, 5]$, correlated significantly ($p = +0.577$) with the true attractiveness score. The proposed method clearly outperforms conventional methods based on distances ($p = +0.296$) and symmetry heuristics ($p = -0.160$).

4. Training Objective & Optimization

FaceRankNet is trained according to a fixed-weighted multi-task objective aimed at maximizing the trade-off between the overall model accuracy and specific organ features. The objective can be defined as follows:

$$L_{total} = L_{reg} + 1.0L_{rank} + 0.01L_{div}$$

- Regression Loss (L_{reg}):** Reduces MSE between global prediction and ground truth ratings.
- Pairwise Ranking Loss (L_{rank}):** Applies a margin-masked logistic loss per organ, preserving the ordinality by considering only those pairs where beauty-axis pseudo-score difference > 0.3 .
- Diversity Regularization (L_{div}):** Attempts to maximize variance across five organ scores in order to avoid the formation of degenerate global predictions, using hinge losses to avoid saturating sigmoid around $[1, 5]$.
- Pair Selection:** Combines the holistic-gap filter (gap > 0.5) with weighted and hard pair selection in order to over-sample extremely beautiful faces in a ratio of $\approx 3.6\times$.

5. Results & Error Structure

When evaluated on the SCUT-FBP5500 dataset (test data set containing 1,100 images), the network produces a PCC of 0.6222 and a MAE of 0.4410 on the test set.

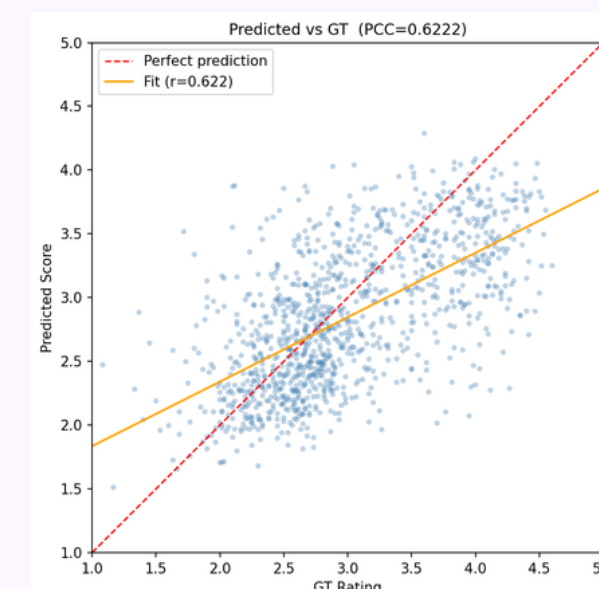


Figure 2: Predicted vs. Ground Truth Scores showcasing overall linear correlation on the test set.

While FaceRankNet successfully generates non-degenerate local scores for each organ, which can be shown to fulfill the positive correlation criterion based on the Spearman measure of validity, the system shows typical regression to the mean errors where extreme low scores are overpredicted by 0.60 and extreme high scores are underpredicted by -0.82, producing score compression whereby the predicted scores tend towards unimodality even though the truth distribution is bimodal.

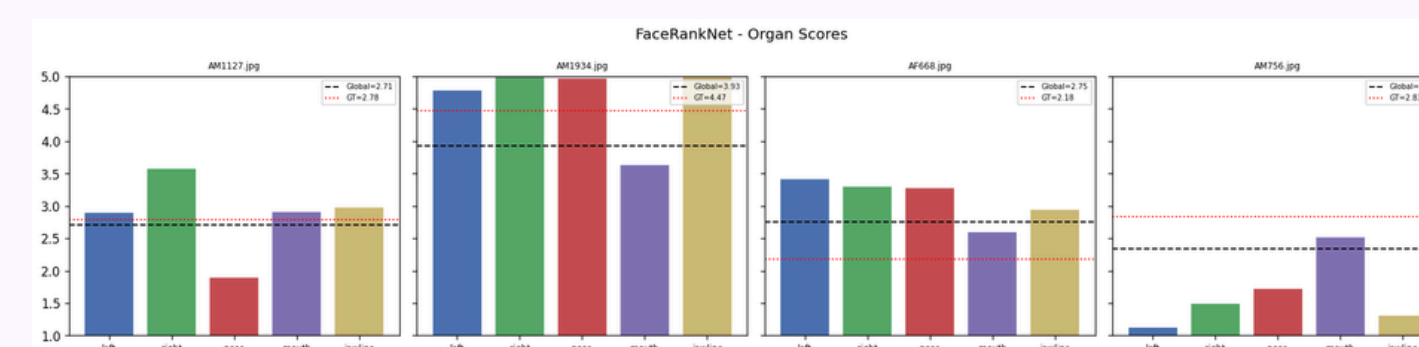


Figure 3: FaceRankNet successfully isolates local organ scores distinct from the holistic evaluation.

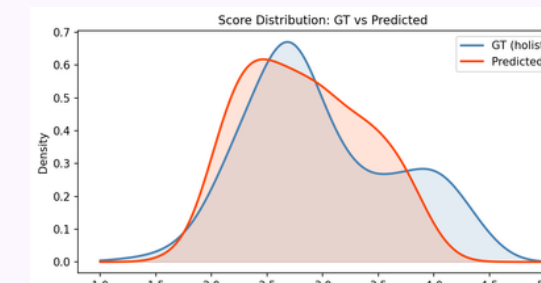


Figure 4: Score compression density plot visualizing the regression-to-the-mean phenomenon.

6. Fairness & Demographic Parity

Without the presence of RGB pixels and through the use of per-ethnicity beauty axis (H1) with symmetric normalizations, the purely geometric design ensures that the demographic bias is structurally addressed, yielding a Demographic Parity Difference (DPD) score of 0.0855 between Asians and Caucasians. On the other hand, the prediction results for the Caucasian group are noisier and less correlated (PCC = 0.454) than those of the Asian group (PCC = 0.672).

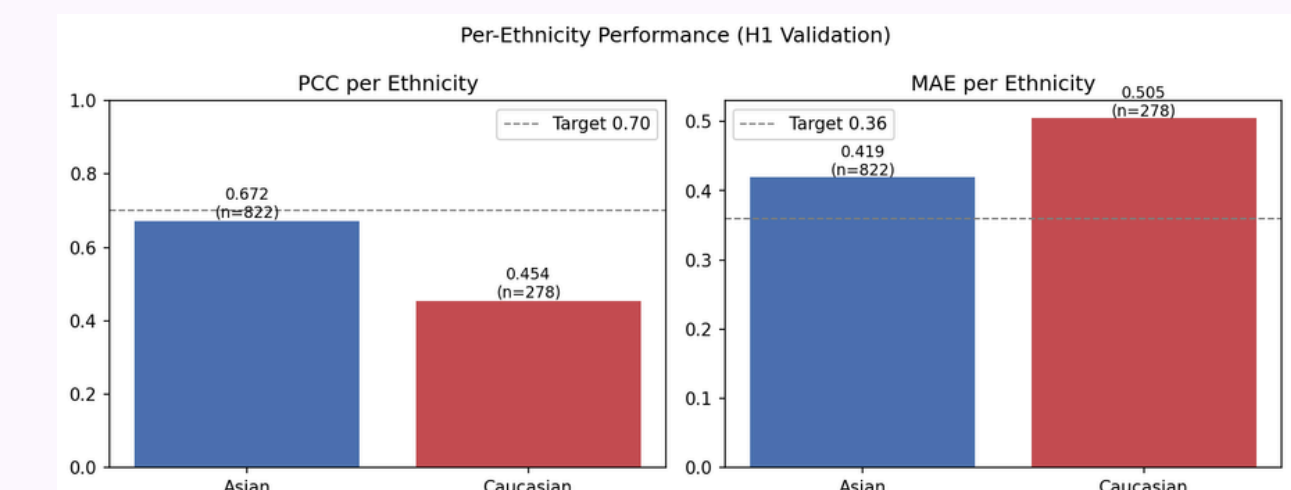


Figure 5: Per-ethnicity PCC and MAE validating the minimal absolute demographic parity difference.

7. Comparison, Future Work & Safety

In line with the benchmarking approaches based on landmarks like GeoFusion-Net (0.612), it is possible through the FaceRankNet model to show how a purely geometric-based method can effectively capture all aesthetic structural information. In order to counteract future score compression, we suggest longer training periods, targeted distributions, and matched quantile pseudo labels to accurately approximate the true shape of the distribution.

With regard to safety and ethics in particular, any kind of automatic aesthetic scoring system must have adequate protection measures against dual use, which include informed consent, source of data, model abstention, and the absolute rejection of minors for participation in the evaluation process. The model interpretability itself will act as a safety feature of the system.

References & QR Code

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